Inverting the K-Function at Infinity

Lambert–W Normalization, All-Orders Transseries,
and a General Theory of Power–Logarithmic Reversion

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Abstract

The K-function is the complex interpolation of the hyperfactorial, $K(n+1) = \prod_{j=1}^{n} j^j$. Its logarithm grows like $\frac{1}{2}x^2 \log x - \frac{1}{4}x^2$, so its large positive inverse is not naturally expanded in ordinary powers of $1/\log Z$. The correct first step is to invert the dominant power–logarithmic phase exactly by the Lambert W-function. After the unique centering $r = x - \frac{1}{2}$, we prove that, with $Y = \log Z$,

$$w = W_0(4Y/\mathrm{e}), \qquad \rho = 2\sqrt{Y/w} = \exp\left(\frac{w+1}{2}\right), \qquad L = \log \rho,$$

the principal inverse has the all-orders transseries

$$K_+^{-1}(Z) \sim \frac{1}{2} + \rho + \sum_{n\geq 1} V_n(L)\rho^{1-2n}.$$

The coefficients V_n are rational Laurent polynomials in L, with polynomial dependence on $C = \log A - \frac{1}{8}$, and are generated by an explicit triangular recurrence. We compute the first coefficients and prove the asymptotic validity of every finite truncation.

A second, substantially faster normal form is obtained from $T = x^2 - x + \frac{1}{6}$. In this coordinate,

$$\log K(x) \sim \frac{1}{4}T(\log T - 1) + C + \sum_{n\geq 1} d_n T^{-n},$$

so the leading inverse is

$$K_+^{-1}(Z) \approx \frac{1}{2} + \sqrt{\frac{1}{12} + \frac{4(Y-C)}{W_0(4(Y-C)/\mathrm{e})}}.$$

An all-orders correction series for T is derived. Numerical tests show that this accelerated formula and its first corrections are highly accurate even at moderate arguments.

The K-function is then used as a model for a general inversion theorem for transseries with dominant part $ax^p(\log x - b)$. We give the exact Lambert–W normalizer, a complete formal recurrence over a slow coefficient field, residual-to-error estimates, Newton acceleration, branch formulas in the complex plane, and the transport rule for exponentially small sectors. The derivations are self-contained at the asymptotic level; no claim of literature priority is made, and the results have not been machine-checked.

Contents

Main result

Let K_+ be the restriction of the K-function to $[2, \infty)$, and let $Z \to +\infty$. Put

$$Y = \log Z, \qquad C = \log A - \frac{1}{8}, \qquad w = W_0\left(\frac{4Y}{\mathrm{e}}\right), \qquad \rho = 2\sqrt{\frac{Y}{w}}, \qquad L = \frac{w+1}{2}.$$

Then

$$\boxed{K_+^{-1}(Z) \sim \frac{1}{2} + \rho + \frac{V_1(L)}{\rho} + \frac{V_2(L)}{\rho^3} + \frac{V_3(L)}{\rho^5} + \cdots}$$

where

$$E = C - \frac{L}{24}, \qquad V_1 = -\frac{E}{L} = \frac{1}{24} - \frac{C}{L},$$
$$V_2 = -\frac{1}{L}\left(\frac{L+1}{2}V_1^2 - \frac{V_1}{24} - \frac{7}{5760}\right),$$

and

$$V_3 = -\frac{1}{L}\left((L+1)V_1V_2 + \frac{V_1^3}{6} - \frac{1}{24}\left(V_2 - \frac{V_1^2}{2}\right) + \frac{7}{2880}V_1 + \frac{31}{161280}\right).$$

All subsequent coefficients are determined uniquely by the recurrence in Theorem 6.1. For computation, the accelerated coordinate

$$T = x^2 - x + \frac{1}{6}$$

is preferable. With

$$\eta = Y - C, \qquad \omega = W_0\left(\frac{4\eta}{\mathrm{e}}\right), \qquad \tau = \frac{4\eta}{\omega}, \qquad S = \log\tau = \omega + 1,$$

one has

$$\boxed{K_+^{-1}(Z) \sim \frac{1}{2} + \sqrt{\frac{1}{12} + \tau\left(1 + \sum_{n\geq 2} U_n(S)\tau^{-n}\right)}}$$

with

$$U_2 = \frac{1}{120S}, \qquad U_3 = -\frac{31}{22680S},$$
$$U_4 = \frac{1030S^2 - 126S - 63}{1814400S^3}, \qquad U_5 = -\frac{16110S^2 - 1023S - 341}{29937600S^3}.$$

1 Introduction and orientation

1.1 Why this inverse requires a transseries

The hyperfactorial

$$H(n) = \prod_{j=1}^{n} j^j$$

grows faster than an ordinary exponential but slower than a function such as $\exp(n^3)$. Its continuous interpolation by the K-function satisfies

$$\log K(x) = \frac{1}{2}x^2\log x - \frac{1}{4}x^2 + \text{lower-order terms}.$$

Consequently, solving $K(x) = Z$ begins with the nonlinear balance

$$\frac{1}{2}x^2 \log x \asymp \log Z.$$

This is already outside the scope of a pure power series in $1/\log Z$. The leading scale contains a Lambert W-function; expansion of that W-function introduces the iterated logarithms

$$\log\log Z, \qquad \log\log\log Z,$$

and the lower-order terms of $\log K$ generate additional algebraic sectors in inverse powers of the Lambert-normalized variable. These scales are strictly ordered but not by a single small parameter. That is precisely the setting in which a transseries is the natural language.

1.2 The central strategy

The development has four layers.

1. Center the independent variable at $x = \frac{1}{2}$. This eliminates an otherwise troublesome $r \log r$ term and leaves an even inverse-power expansion.
2. Invert the dominant phase exactly with W_0. This produces a large reference scale ρ and a slow variable $L = \log \rho$.
3. Solve the remaining equation in the complete ring of formal powers of $q = \rho^{-2}$, with coefficients depending slowly on L. The linearized coefficient is L, so the recurrence is triangular.
4. Improve the coordinate further by introducing $T = x^2 - x + \frac{1}{6}$. This absorbs the logarithmic correction into the leading phase and delays the first unresolved term by a full asymptotic sector.

The same mechanism works for a broad class of maps with dominant part $ax^p(\log x - b)$. The K-function is especially instructive because both normalizations can be carried out in closed form and because the Bernoulli structure of its direct expansion is explicit.

1.3 Relation to the accompanying transseries program

The series-and-transseries directory of the ProveIt project is devoted to the algebra, composition, and reversion of asymptotic expansions rather than to a single special function [9]. The present article follows that operational viewpoint: the K-function supplies a concrete test case, while the later sections isolate the reusable inversion calculus. In particular, exact inversion of the dominant monomial is performed before any formal reversion, and analytic error estimates are separated from formal coefficient algebra.

1.4 Status and scope

The positive real inverse is treated completely to all algebraic orders. A sectorial complex version is also given, but a global classification of the Riemann surface of the inverse K-function is not attempted. Exponentially small terms are discussed through a general transport theorem and through the known exponentially improved theory of the Barnes G-function [7]; explicit Stokes multipliers for every inverse branch remain a separate problem.

The coefficient recurrences, the centered inverse expansion, and the accelerated T-coordinate expansion were derived for this report and were checked symbolically and numerically. No priority claim is made: the literature on hyperfactorial asymptotics is extensive, including refinements such as [11], and an exhaustive historical search is beyond the scope of this article.

2 The K-function and its principal inverse

2.1 Definitions and elementary identities

For positive real x, one convenient definition is

$$K(x) = (2\pi)^{-(x-1)/2} \exp\left\{\binom{x}{2} + \int_0^{x-1} \log\Gamma(t+1)\, \mathrm{d}t\right\}. \tag{2.1}$$

Equivalent complex definitions may be written in terms of the Hurwitz zeta function or the Barnes G-function:

$$K(z) = \exp(\zeta'(-1, z) - \zeta'(-1)), \tag{2.2}$$

and, on compatible logarithmic branches,

$$K(z)G(z) = \Gamma(z)^{z-1}. \tag{2.3}$$

These standard identities are reviewed in [6, 4]. The functional equation is

$$K(z+1) = z^z K(z), \tag{2.4}$$

so that

$$K(n+1) = \prod_{j=1}^{n} j^j. \tag{2.5}$$

Throughout the positive-real analysis we write

$$\kappa(x) = \log K(x),$$

with the ordinary real logarithm.

Lemma 2.1 (Logarithmic derivatives)**.** *For $x > 0$,*

$$\kappa'(x) = \log\Gamma(x) + x - \frac{1}{2} - \frac{1}{2}\log(2\pi), \tag{2.6}$$

$$\kappa''(x) = \psi(x) + 1. \tag{2.7}$$

Proof. Differentiate Equation (2.1). The derivative of the upper limit contributes $\log\Gamma(x)$, the quadratic term contributes $x - \frac{1}{2}$, and the prefactor contributes $-\frac{1}{2}\log(2\pi)$. Differentiating once more gives Equation (2.7). □

Theorem 2.2 (The principal positive inverse)**.** *The restriction*

$$K_+ = K|_{[2,\infty)} : [2, \infty) \longrightarrow [1, \infty)$$

is strictly increasing and onto. Hence it possesses a unique inverse

$$K_+^{-1} : [1, \infty) \longrightarrow [2, \infty).$$

Proof. Since ψ is increasing on $(0, \infty)$ and $\psi(2) = 1 - \gamma > 0$, Equation (2.7) gives $\kappa''(x) > 0$ for $x \geq 2$. Also

$$\kappa'(2) = \frac{3}{2} - \frac{1}{2}\log(2\pi) > 0.$$

Thus κ', and therefore K', is positive on $[2, \infty)$. Equation (2.5) gives $K(2) = 1$, while the asymptotic formula proved below implies $K(x) \to \infty$. The conclusion follows. □

Remark 2.3. The K-function is not monotone on all of $(0, \infty)$; the restriction in Theorem 2.2 selects the branch that tends to infinity and is the branch relevant to large arguments.

2.2 The Glaisher–Kinkelin constant

The constant appearing in the asymptotic expansion is

$$A = \exp\left(\frac{1}{12} - \zeta'(-1)\right) = 1.2824271291006226368\ldots, \tag{2.8}$$

and we shall repeatedly use

$$C = \log A - \frac{1}{8} = 0.12375447703378426255\ldots. \tag{2.9}$$

3 The direct large-argument expansion of $\log K$

3.1 An all-orders formula

We use Bernoulli numbers in the convention

$$\frac{t}{\mathrm{e}^t - 1} = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!}.$$

Theorem 3.1 (Direct K-function expansion)**.** *For every fixed integer $N \geq 1$, as $x \to +\infty$,*

$$\begin{aligned}\kappa(x) = {} & \left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{12}\right)\log x - \frac{x^2}{4} + \log A \\ & - \sum_{n=1}^{N-1} \frac{B_{2n+2}}{(2n)(2n+1)(2n+2)x^{2n}} + O(x^{-2N}).\end{aligned} \tag{3.1}$$

The same expansion holds uniformly in closed subsectors $|\arg x| \leq \pi - \delta$, with a fixed branch of $\operatorname{Log} x$.

Proof. From Equation (2.6) and Stirling's expansion,

$$\log\Gamma(x) = \left(x - \frac{1}{2}\right)\log x - x + \frac{1}{2}\log(2\pi) + \sum_{m=1}^{N} \frac{B_{2m}}{(2m)(2m-1)x^{2m-1}} + O(x^{-2N-1}),$$

we obtain

$$\kappa'(x) = \left(x - \frac{1}{2}\right)\log x - \frac{1}{2} + \sum_{m=1}^{N} \frac{B_{2m}}{(2m)(2m-1)x^{2m-1}} + O(x^{-2N-1}). \tag{3.2}$$

Termwise integration yields Equation (3.1), apart from an integration constant. That constant is $\log A$, as follows either from the classical hyperfactorial asymptotic at integers or from the Barnes G-function relation Equation (2.3) and the standard asymptotic normalization of G; see [4]. Uniform sectorial remainder bounds follow from the corresponding sectorial Stirling expansion and integration along a ray inside the sector. □

The first terms are

$$\begin{aligned}\kappa(x) \sim {} & \left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{12}\right)\log x - \frac{x^2}{4} + \log A + \frac{1}{720x^2} - \frac{1}{5040x^4} \\ & + \frac{1}{10080x^6} - \frac{1}{9504x^8} + \frac{691}{3603600x^{10}} - \cdots.\end{aligned} \tag{3.3}$$

3.2 Divergence and the least-term scale

The Bernoulli numbers satisfy

$$|B_{2n}| \sim \frac{2(2n)!}{(2\pi)^{2n}}.$$

Consequently, Equation (3.1) is normally divergent. Its smallest term occurs when the index is proportional to πx, and the optimally truncated remainder has an exponential scale comparable to $\mathrm{e}^{-2\pi x}$. This does not invalidate the algebraic expansion: each fixed finite truncation is a valid Poincare approximation. It does mean that an exponentially improved analysis must eventually append additional sectorial data, a point revisited in Section 12.

4 The unique centered normal form

4.1 Why the shift by one half is forced

Let $x = r + h$ in the leading part of Equation (3.1). The coefficient of $r \log r$ is $h - \frac{1}{2}$. Hence

$$h = \frac{1}{2}$$

is the unique affine shift that removes this term. This is more than a cosmetic simplification: after the shift, all inverse-power corrections are even, and the inverse series advances in steps of r^{-2}.

Put

$$r = x - \frac{1}{2}. \tag{4.1}$$

We use Bernoulli polynomials defined by

$$\frac{t\mathrm{e}^{ut}}{\mathrm{e}^{t} - 1} = \sum_{n=0}^{\infty} B_n(u) \frac{t^n}{n!}.$$

Recall

$$B_n\left(\frac{1}{2}\right) = \left(2^{1-n} - 1\right) B_n.$$

Theorem 4.1 (Centered even expansion)**.** *As $r \to +\infty$,*

$$\kappa\left(r + \frac{1}{2}\right) \sim \left(\frac{r^2}{2} - \frac{1}{24}\right) \log r - \frac{r^2}{4} + C + \sum_{n=1}^{\infty} c_n r^{-2n}, \tag{4.2}$$

where

$$c_n = -\frac{B_{2n+2}(1/2)}{(2n)(2n+1)(2n+2)}. \tag{4.3}$$

For every fixed N, truncation after $c_{N-1} r^{-2N+2}$ has remainder $O(r^{-2N})$.

Proof. The shifted Stirling expansion is

$$\log \Gamma\left(r + \frac{1}{2}\right) \sim r \log r - r + \frac{1}{2} \log(2\pi) + \sum_{m=1}^{\infty} \frac{B_{2m}(1/2)}{(2m)(2m-1) r^{2m-1}}. \tag{4.4}$$

Substitution into Equation (2.6) gives

$$\frac{\mathrm{d}}{\mathrm{d}r} \kappa\left(r + \frac{1}{2}\right) \sim r \log r - \frac{1}{24r} + \sum_{n=1}^{\infty} \frac{B_{2n+2}(1/2)}{(2n+1)(2n+2) r^{2n+1}}.$$

Integrating term by term yields Equation (4.2) and Equation (4.3). Substitution of $x = r + \frac{1}{2}$ into Equation (3.1) fixes the constant as $C = \log A - \frac{1}{8}$. □

The first coefficients are

$$\begin{aligned} c_1 &= -\frac{7}{5760}, & c_2 &= \frac{31}{161280}, & c_3 &= -\frac{127}{1290240}, \\ c_4 &= \frac{511}{4866048}, & c_5 &= -\frac{1414477}{7380172800}, & c_6 &= \frac{8191}{15335424}. \end{aligned} \tag{4.5}$$

4.2 Separation of the dominant phase

Define

$$\Phi(r) = \frac{1}{2}r^2 \log r - \frac{1}{4}r^2. \tag{4.6}$$

Then Equation (4.2) becomes

$$\kappa\left(r + \frac{1}{2}\right) \sim \Phi(r) - \frac{1}{24}\log r + C + \sum_{n\geq 1} c_n r^{-2n}. \tag{4.7}$$

This form makes the hierarchy transparent:

$$\Phi(r) \asymp r^2 \log r, \qquad -\frac{1}{24}\log r + C = O(\log r), \qquad c_n r^{-2n} = O(r^{-2n}).$$

5 Exact inversion of the dominant phase

5.1 Lambert W appears without approximation

Let

$$Y = \log Z. \tag{5.1}$$

The dominant equation is

$$\Phi(\rho) = Y. \tag{5.2}$$

Set $w = 2\log\rho - 1$. Since $\rho^2 = \mathrm{e}^{w+1}$,

$$Y = \frac{1}{4}\rho^2(2\log\rho - 1) = \frac{\mathrm{e}}{4} w \mathrm{e}^{w}.$$

Thus

$$w = W_0\left(\frac{4Y}{\mathrm{e}}\right), \tag{5.3}$$

and

$$\rho = \exp\left(\frac{w+1}{2}\right) = 2\sqrt{\frac{Y}{w}}, \qquad L = \log\rho = \frac{w+1}{2}. \tag{5.4}$$

No asymptotic approximation has yet been made: ρ is the exact positive solution of Equation (5.2).

The zeroth approximation to the inverse K-function is therefore

$$K_+^{-1}(Z) = \frac{1}{2} + \rho + o(\rho). \tag{5.5}$$

5.2 Elementary iterated-log expansion

The W-adapted formula Equation (5.4) is the cleanest representation. If an expansion using only elementary logarithms is desired, set

$$U = \log\left(\frac{4Y}{\mathrm{e}}\right), \qquad V = \log U. \tag{5.6}$$

The standard large-argument expansion of W_0 [3, 5] gives

$$\rho = 2\sqrt{\frac{Y}{U}}\left[1 + \frac{V}{2U} + \frac{V(3V-4)}{8U^2} + \frac{V(5V^2-16V+8)}{16U^3} \right.$$
$$\left. + \frac{V(105V^3 - 568V^2 + 720V - 192)}{384U^4} + O\left(\frac{V^5}{U^5}\right)\right]. \tag{5.7}$$

Thus the leading sector alone contains arbitrarily deep powers of $1/\log Y$, with polynomial coefficients in $\log\log Y$.

Important ordering point

The additive constant $\frac{1}{2}$ in $K_+^{-1}(Z)$ is smaller than every fixed truncation error of the elementary expansion Equation (5.7): for each fixed N,

$$\frac{\rho}{U^N} \longrightarrow \infty.$$

Hence the correct transseries order is:

$$\text{complete Lambert/iterated-log sector of size } \rho, \qquad \frac{1}{2}, \qquad \rho^{-1}, \qquad \rho^{-3}, \ldots.$$

Writing a few inverse-log terms and then appending $\frac{1}{2}$ does not produce an asymptotically ordered one-dimensional series.

6 The canonical centered inverse transseries

6.1 The normalized implicit equation

Put

$$q = \rho^{-2}, \qquad r = \rho(1+v), \qquad v = \sum_{n\geq 1} V_n(L) q^n. \tag{6.1}$$

The exact dominant identity $\Phi(\rho) = Y$ suggests normalizing the remaining equation by ρ^2. Define

$$\mathcal{D}_L(v) = \frac{1}{2}(1+v)^2\big(L + \log(1+v)\big) - \frac{1}{4}(1+v)^2 - \frac{1}{2}L + \frac{1}{4}. \tag{6.2}$$

Then

$$\mathcal{D}_L(v) = Lv + \frac{L+1}{2}v^2 + \sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k(k-1)(k-2)} v^k. \tag{6.3}$$

Let

$$E = E(L) = C - \frac{L}{24}. \tag{6.4}$$

Substitution of Equation (6.1) into Equation (4.7) gives the formal implicit equation

$$\boxed{\mathcal{D}_L(v) + q\left[E - \frac{1}{24}\log(1+v) + \sum_{m\geq 1} c_m q^m (1+v)^{-2m}\right] = 0.} \tag{6.5}$$

6.2 Formal existence, uniqueness, and recurrence

Let

$$\mathscr{R} = \mathbb{Q}[C, L, L^{-1}]$$

and consider the complete q-adic ring $\mathscr{R}[[q]]$. The variable L is treated as a slow coefficient, while q records the genuinely small algebraic scale.

Theorem 6.1 (Canonical all-orders recurrence)**.** *Equation (6.5) has a unique solution*

$$v \in q\mathscr{R}[[q]].$$

Writing

$$v = \sum_{n\geq 1} V_n(L)q^n,$$

the coefficients are obtained recursively as follows. Let

$$v_{<n} = \sum_{j=1}^{n-1} V_j(L)q^j.$$

Then

$$V_n(L) = -\frac{1}{L}[q^n]\Bigg\{\mathcal{D}_L(v_{<n}) + q\Bigg[E - \frac{1}{24}\log(1 + v_{<n}) + \sum_{m\geq 1} c_m q^m (1 + v_{<n})^{-2m}\Bigg]\Bigg\}. \tag{6.6}$$

Only $c_1, \ldots, c_{n-1}$ can contribute to V_n.

Proof. Let the left side of Equation (6.5) be $\mathcal{F}(v, q)$. Then

$$\mathcal{F}(0,0) = 0, \qquad \partial_v \mathcal{F}(0,0) = L.$$

Since L is a unit in $\mathscr{R}$, the formal implicit-function theorem in the complete q-adic ring gives a unique solution in $q\mathscr{R}[[q]]$. More concretely, suppose $V_1, \ldots, V_{n-1}$ are known. At order q^n, the only occurrence of the new coefficient is the linear term LV_nq^n from $\mathcal{D}_L(v)$. All other contributions depend only on earlier coefficients. Solving the coefficient equation gives Equation (6.6). □

6.3 The first coefficients

The first three coefficients are

$$V_1 = -\frac{E}{L} = \frac{1}{24} - \frac{C}{L}, \tag{6.7}$$

$$V_2 = -\frac{1}{L}\left(\frac{L+1}{2}V_1^2 - \frac{1}{24}V_1 + c_1\right), \qquad c_1 = -\frac{7}{5760}, \tag{6.8}$$

and

$$V_3 = -\frac{1}{L}\left((L+1)V_1V_2 + \frac{1}{6}V_1^3 - \frac{1}{24}\left(V_2 - \frac{1}{2}V_1^2\right) - 2c_1V_1 + c_2\right), \tag{6.9}$$

where $c_2 = 31/161280$. An equivalent closed expression for V_2 is

$$V_2 = \frac{7L^2 - 240EL - 2880(L+1)E^2}{5760L^3}. \tag{6.10}$$

For reference, the next recursion is

$$V_4 = -\frac{1}{L}\Bigg[(L+1)\left(V_1V_3 + \frac{1}{2}V_2^2\right) + \frac{1}{2}V_1^2V_2 - \frac{1}{24}V_1^4 \\ - \frac{1}{24}\left(V_3 - V_1V_2 + \frac{1}{3}V_1^3\right) + c_1(-2V_2 + 3V_1^2) - 4c_2V_1 + c_3\Bigg]. \tag{6.11}$$

As $L \to \infty$, the coefficients remain bounded; for example,

$$\begin{aligned} V_1 &= \frac{1}{24} - \frac{C}{L}, \\ V_2 &= -\frac{1}{1152} + \frac{C/24 + 1/480}{L} - \frac{C^2}{2L^2} + O(L^{-3}), \\ V_3 &= \frac{1}{27648} - \frac{C/384 + 311/725760}{L} + O(L^{-2}). \end{aligned}$$

6.4 The inverse expansion and its analytic meaning

Theorem 6.2 (Principal inverse transseries)**.** *Let $Z \to +\infty$, and define Y, w, ρ, L by Equations (5.1), (5.3) and (5.4). Then*

$$\boxed{K_+^{-1}(Z) \sim \frac{1}{2} + \rho + \sum_{n=1}^{\infty} V_n(L)\rho^{1-2n},} \tag{6.12}$$

where the V_n are given by Equation (6.6). More precisely, for every fixed $N \geq 1$,

$$K_+^{-1}(Z) - \left[\frac{1}{2} + \rho + \sum_{n=1}^{N-1} V_n(L)\rho^{1-2n}\right] = O\left(\rho^{1-2N}\right). \tag{6.13}$$

Proof. Formal substitution into Equation (4.7) gives a residual whose normalized expansion vanishes through the prescribed power of $q = \rho^{-2}$. The direct asymptotic expansion of κ may be taken to sufficiently many terms to make its own remainder smaller than the first omitted inverse term. On the positive ray,

$$\kappa'\left(r + \frac{1}{2}\right) = r \log r + O(r^{-1}) \asymp \rho L. \tag{6.14}$$

The mean-value theorem therefore converts the residual into an error in r one derivative-scale smaller. The first omitted formal term is $V_N(L)\rho^{1-2N}$, and $V_N(L) = O(1)$ for fixed N. This proves Equation (6.13). □

Practical formula

The first three useful approximants are

$$\begin{aligned} X_0(Z) &= \frac{1}{2} + \rho, \\ X_1(Z) &= X_0(Z) + \frac{V_1(L)}{\rho}, \\ X_2(Z) &= X_1(Z) + \frac{V_2(L)}{\rho^3}, \\ X_3(Z) &= X_2(Z) + \frac{V_3(L)}{\rho^5}. \end{aligned}$$

They require only W_0, elementary arithmetic, and the constant C. One Newton correction using the exact K-function generally improves the accuracy far beyond the displayed truncation.

7 A faster quadratic normal form

7.1 A general logarithmic-absorption lemma

The centered K expansion contains the slow correction $-\frac{1}{24}\log r$. It can be absorbed into a new algebraic coordinate. The following elementary observation explains why.

Lemma 7.1 (Absorbing a logarithmic correction)**.** *Suppose*

$$F(r) = ar^p(\log r - b) + \beta \log r + \gamma + O(r^{-p}), \qquad a \neq 0, \tag{7.1}$$

and set

$$T = r^p + \alpha, \qquad \Psi(T) = \frac{a}{p}T(\log T - pb). \tag{7.2}$$

Then

$$\Psi(r^p + \alpha) = ar^p(\log r - b) + a\alpha \log r + a\alpha\left(\frac{1}{p} - b\right) + O(r^{-p}). \tag{7.3}$$

Thus the unique constant shift that absorbs $\beta \log r$ is

$$\alpha = \frac{\beta}{a}. \tag{7.4}$$

Proof. Insert $T = r^p + \alpha$ into Equation (7.2) and use

$$\log T = p\log r + \log(1 + \alpha r^{-p}) = p\log r + \alpha r^{-p} + O(r^{-2p}).$$

Collecting the terms of orders $r^p \log r$, r^p, $\log r$, and 1 gives Equation (7.3). □

For the K-function,

$$a = \frac{1}{2}, \qquad p = 2, \qquad b = \frac{1}{2}, \qquad \beta = -\frac{1}{24}.$$

Hence $\alpha = -\frac{1}{12}$, and the natural coordinate is

$$T = r^2 - \frac{1}{12} = x^2 - x + \frac{1}{6}. \tag{7.5}$$

The positive branch is recovered from

$$x = \frac{1}{2} + \sqrt{T + \frac{1}{12}}. \tag{7.6}$$

7.2 The accelerated direct expansion

Define

$$\Psi(T) = \frac{1}{4}T(\log T - 1). \tag{7.7}$$

Theorem 7.2 (Quadratic normal form)**.** *As $T \to +\infty$,*

$$\kappa(x) \sim \Psi(T) + C + \sum_{n=1}^{\infty} d_n T^{-n}, \qquad T = x^2 - x + \frac{1}{6}, \tag{7.8}$$

where, with $a_0 = 1/12$,

$$d_n = \frac{(-1)^n a_0^{n+1}}{4(n+1)} + \sum_{m=1}^{n} (-1)^{n-m} \binom{n-1}{m-1} a_0^{n-m} c_m. \tag{7.9}$$

The first coefficients are

$$\begin{aligned} d_1 &= -\frac{1}{480}, & d_2 &= \frac{31}{90720}, & d_3 &= -\frac{103}{725760}, \\ d_4 &= \frac{179}{1330560}, & d_5 &= -\frac{6480899}{28021593600}, & d_6 &= \frac{3485299}{5604318720}. \end{aligned} \tag{7.10}$$

Proof. Because $r^2 = T + a_0$ and $\frac{1}{2}r^2 - \frac{1}{24} = T/2$, the first two terms of Equation (4.2) satisfy the exact identity

$$\begin{aligned} \left(\frac{r^2}{2} - \frac{1}{24}\right)\log r - \frac{r^2}{4} &= \frac{T}{4}\log(T + a_0) - \frac{T}{4} - \frac{a_0}{4} \\ &= \Psi(T) + \frac{T}{4}\log\left(1 + \frac{a_0}{T}\right) - \frac{a_0}{4}. \end{aligned}$$

The term with first power of a_0/T cancels $a_0/4$. Expanding the remaining logarithm and

$$(T + a_0)^{-m} = T^{-m}(1 + a_0/T)^{-m}$$

then collecting the coefficient of T^{-n} gives Equation (7.9). □

The importance of Equation (7.8) is structural. The correction of size $\log r$, which produced a displacement of size r^{-1} in the canonical inverse, has disappeared. The first unresolved direct term is now d_1/T, and it produces an inverse displacement only of order $T^{-3/2}/\log T$.

7.3 Exact leading inversion in the T-coordinate

Set

$$\eta = Y - C. \tag{7.11}$$

The dominant equation $\Psi(\tau) = \eta$ is

$$\frac{1}{4}\tau(\log\tau - 1) = \eta.$$

Let

$$\omega = W_0\left(\frac{4\eta}{\mathrm{e}}\right), \qquad \tau = \frac{4\eta}{\omega} = \mathrm{e}^{\omega+1}, \qquad S = \log\tau = \omega + 1. \tag{7.12}$$

Then $\Psi(\tau) = \eta$ exactly. The leading accelerated approximation is therefore

$$\boxed{X_{\mathrm{acc},0}(Z) = \frac{1}{2} + \sqrt{\frac{1}{12} + \frac{4(\log Z - C)}{W_0(4(\log Z - C)/\mathrm{e})}}.} \tag{7.13}$$

7.4 All-orders accelerated recurrence

Write

$$T = \tau(1 + u), \qquad \nu = \tau^{-1}, \qquad \nu \to 0^+. \tag{7.14}$$

Define

$$\mathcal{E}_S(u) = (1 + u)(S + \log(1 + u) - 1) - (S - 1). \tag{7.15}$$

Its expansion is

$$\mathcal{E}_S(u) = Su + \sum_{k=2}^{\infty} \frac{(-1)^k}{k(k-1)}u^k. \tag{7.16}$$

After multiplying the equation by $4/\tau$, the normalized implicit relation becomes

$$\boxed{\mathcal{E}_S(u) + 4\sum_{m\geq 1} d_m \nu^{m+1}(1 + u)^{-m} = 0.} \tag{7.17}$$

Since the perturbation starts at ν^2, so does u.

Theorem 7.3 (Accelerated all-orders recurrence)**.** *Equation (7.17) has a unique solution*

$$u = \sum_{n=2}^{\infty} U_n(S)v^n \quad \textit{in} \quad v^2\mathbb{Q}[S, S^{-1}][[v]]. \tag{7.18}$$

For

$$u_{<n} = \sum_{j=2}^{n-1} U_j(S)v^j,$$

the triangular recurrence is

$$U_n(S) = -\frac{1}{S}[v^n]\left\{\mathcal{E}_S(u_{<n}) + 4\sum_{m\geq 1} d_m v^{m+1}(1+u_{<n})^{-m}\right\}. \tag{7.19}$$

The first coefficients are

$$U_2 = \frac{1}{120S}, \qquad U_3 = -\frac{31}{22680S}, \tag{7.20}$$

$$U_4 = \frac{1030S^2 - 126S - 63}{1814400S^3}, \qquad U_5 = -\frac{16110S^2 - 1023S - 341}{29937600S^3}. \tag{7.21}$$

Proof. The left side of Equation (7.17) vanishes at $(u, v) = (0, 0)$, and its derivative with respect to u there is S, a unit in the chosen coefficient ring. Formal implicit inversion therefore gives a unique series. Since the forcing term begins with $4d_1v^2$, the coefficients of v^0 and v^1 vanish and $u = O(v^2)$. Coefficient extraction gives Equation (7.19). Direct calculation gives Equations (7.20) and (7.21). □

Corollary 7.4 (Accelerated inverse transseries)**.** *As $Z \to +\infty$,*

$$\boxed{K_+^{-1}(Z) \sim \frac{1}{2} + \sqrt{\frac{1}{12} + \tau\left(1 + \sum_{n=2}^{\infty} U_n(S)\tau^{-n}\right)},} \tag{7.22}$$

where τ, S are given by Equation (7.12). If the sum is truncated after $U_N\tau^{-N}$, the error in x is

$$O\left(\tau^{-N-1/2}\right) \qquad (N \geq 2), \tag{7.23}$$

with slow powers of S^{-1} included in the coefficient.

The first two corrections may be displayed without expanding the square root. If

$$R = \sqrt{\tau + \frac{1}{12}},$$

then

$$K_+^{-1}(Z) = \frac{1}{2} + R + \frac{1}{240S\tau R} - \frac{31}{45360S\tau^2 R} + O\left(\tau^{-7/2}\right). \tag{7.24}$$

7.5 The continuous hyperfactorial inverse

Define the continuous hyperfactorial interpolation

$$\mathcal{H}(t) = K(t+1), \qquad t \geq 1.$$

Then

$$\mathcal{H}^{-1}(Z) = K_+^{-1}(Z) - 1. \tag{7.25}$$

In the accelerated coordinate,

$$T = t^2 + t + \frac{1}{6},$$

so Equation (7.22) immediately gives

$$\mathcal{H}^{-1}(Z) \sim -\frac{1}{2} + \sqrt{\frac{1}{12} + \tau\left(1 + \sum_{n=2}^{\infty} U_n(S)\tau^{-n}\right)}. \tag{7.26}$$

This explains the naturally occurring quadratic combination $t^2 + t + \frac{1}{6}$ in refined hyperfactorial asymptotics.

8 Numerical verification

8.1 Reference computation

For positive real x, the Barnes relation gives the stable identity

$$\kappa(x) = (x-1)\log\Gamma(x) - \log G(x). \tag{8.1}$$

Reference inverse values were obtained by Newton iteration

$$x_{j+1} = x_j - \frac{\kappa(x_j) - Y}{\log\Gamma(x_j) + x_j - \frac{1}{2} - \frac{1}{2}\log(2\pi)}, \tag{8.2}$$

using high-precision implementations of Γ, G, and W_0. The signed errors below are approximation minus reference value.

8.2 Canonical centered expansion

Let X_j denote the canonical approximant through V_j, with $X_0 = \frac{1}{2} + \rho$. The results are:

Table 1: Canonical centered inverse: signed errors.

Y	reference x	$X_0 - x$	$X_1 - x$	$X_2 - x$	$X_3 - x$
10	4.966157945496709	$9.1616267\,10^{-3}$	$1.9348166\,10^{-5}$	$1.3471239\,10^{-7}$	$-6.0356982\,10^{-10}$
100	10.915136726993085	$1.0696633\,10^{-3}$	$-2.0464973\,10^{-7}$	$5.1650691\,10^{-10}$	$-2.6895791\,10^{-12}$
1000	27.284589040600087	$-1.5034726\,10^{-4}$	$-2.1337345\,10^{-8}$	$4.2407143\,10^{-12}$	$-3.0318935\,10^{-15}$
10000	73.181059301767169	$-1.7601786\,10^{-4}$	$-7.9953926\,10^{-10}$	$2.7035317\,10^{-14}$	$-2.3761398\,10^{-18}$
10^6	584.235864411692401	$-3.8094767\,10^{-5}$	$-2.5238008\,10^{-13}$	$6.1276265\,10^{-19}$	$-8.3193819\,10^{-25}$

8.3 Accelerated quadratic expansion

Let A_0 be Equation (7.13), and let A_j for $j = 2, 3, 4, 5$ include the terms through U_j inside the square root.

Table 2: Accelerated inverse: signed errors.

Y	$A_0 - x$	$A_2 - x$	$A_3 - x$	$A_4 - x$	$A_5 - x$
10	$-1.5587271\,10^{-5}$	$1.2729709\,10^{-7}$	$-2.4670503\,10^{-9}$	$1.1659381\,10^{-10}$	$-9.8337554\,10^{-12}$
100	$-7.8648990\,10^{-7}$	$1.1875374\,10^{-9}$	$-4.3964091\,10^{-12}$	$3.8746388\,10^{-14}$	$-6.1491072\,10^{-16}$
1000	$-3.2972642\,10^{-8}$	$7.5368039\,10^{-12}$	$-4.2730563\,10^{-15}$	$5.6974464\,10^{-18}$	$-1.3723490\,10^{-20}$
10000	$-1.2659858\,10^{-9}$	$3.9307274\,10^{-14}$	$-3.0434681\,10^{-18}$	$5.5017139\,10^{-22}$	$-1.7988347\,10^{-25}$
10^6	$-1.6444009\,10^{-12}$	$7.9154321\,10^{-19}$	$-9.5514849\,10^{-25}$	$2.6701788\,10^{-30}$	$-1.3514777\,10^{-35}$

The accelerated leading approximation already outperforms the first corrected canonical formula over most of the table. This is not accidental: it has exactly resummed the constant C, the logarithmic correction, and the algebraic effects induced by the quadratic change of variable.

9 A general theory of power–logarithmic transseries inversion

9.1 The model class

Let $p > 0$, $\sigma > 0$, and $a \neq 0$. Consider a sectorial asymptotic expansion of the form

$$F(x) \sim ax^p(\log x - b) + \sum_{m=0}^{\infty} x^{p-(m+1)\sigma} A_m(\log x). \tag{9.1}$$

The functions A_m belong to a slow coefficient field $\mathcal{S}$. Typical choices are rational functions of $\log x$, polynomial-logarithmic transseries in deeper logarithms, or sectorial analytic functions admitting such expansions. The essential requirement is that the coefficient field be closed under differentiation, composition with $\log(1+v)$, and inversion of the linearized factor introduced below.

The K-function in the centered variable belongs to this class with

$$a = \frac{1}{2}, \qquad p = 2, \qquad b = \frac{1}{2}, \qquad \sigma = 2, \tag{9.2}$$

and

$$A_0(L) = C - \frac{L}{24}, \qquad A_m(L) = c_m \quad (m \geq 1).$$

9.2 Exact dominant inversion

Theorem 9.1 (Lambert–W normalizer)**.** *Fix a branch W_k and compatible branches of the logarithm and p-th root. The equation*

$$y = a\rho^p(\log\rho - b) \tag{9.3}$$

has the solution

$$\Omega = W_k\left(\frac{p}{a}\mathrm{e}^{-pb}y\right), \tag{9.4}$$

$$\Lambda = \log\rho = b + \frac{\Omega}{p}, \tag{9.5}$$

$$\rho = \exp\left(b + \frac{\Omega}{p}\right) = \left[\frac{py/a}{W_k((p/a)\mathrm{e}^{-pb}y)}\right]^{1/p}. \tag{9.6}$$

Proof. Put $\Omega = p(\log\rho - b)$. Then

$$\rho^p = \mathrm{e}^{pb+\Omega},$$

and Equation (9.3) becomes

$$\Omega\mathrm{e}^{\Omega} = \frac{p}{a}\mathrm{e}^{-pb}y.$$

Applying the selected W branch gives Equation (9.4); the remaining formulas follow immediately. □

9.3 The normalized equation and formal theorem

Set

$$q = \rho^{-\sigma}, \qquad x = \rho(1+v). \tag{9.7}$$

Define

$$\mathcal{D}_{p,\Lambda,b}(v) = (1+v)^p\big(\Lambda - b + \log(1+v)\big) - (\Lambda - b). \tag{9.8}$$

After division by $a\rho^p$, the inversion equation is

$$0 = \mathcal{D}_{p,\Lambda,b}(v) + \frac{1}{a}\sum_{m\geq 0} q^{m+1}(1+v)^{p-(m+1)\sigma}$$

$$\times A_m\big(\Lambda + \log(1+v)\big). \tag{9.9}$$

The linear coefficient is

$$\partial_v \mathcal{D}_{p,\Lambda,b}(0) = p(\Lambda - b) + 1 = \Omega + 1. \tag{9.10}$$

Theorem 9.2 (Formal inversion over a slow coefficient field)**.** *Assume $\Omega + 1$ is invertible in the slow coefficient field $\mathcal{S}$. Then Equation (9.9) has a unique solution*

$$v = \sum_{n \geq 1} V_n q^n \in q\mathcal{S}[[q]]. \tag{9.11}$$

If $v_{<n} = \sum_{j=1}^{n-1} V_j q^j$, then

$$\begin{aligned} V_n = -\frac{1}{\Omega+1}[q^n]\Bigg\{ &\mathcal{D}_{p,\Lambda,b}(v_{<n}) \\ &+ \frac{1}{a}\sum_{m \geq 0} q^{m+1}(1+v_{<n})^{p-(m+1)\sigma} \\ &\qquad \times A_m\big(\Lambda + \log(1+v_{<n})\big)\Bigg\}. \end{aligned} \tag{9.12}$$

Consequently,

$$F^{-1}(y) \sim \rho(y)\left(1 + \sum_{n \geq 1} V_n(y)\rho(y)^{-n\sigma}\right). \tag{9.13}$$

Proof. The left side of Equation (9.9) vanishes at $(v, q) = (0, 0)$, and by Equation (9.10) its derivative with respect to v is the unit $\Omega + 1$. The formal implicit-function theorem gives existence and uniqueness in the complete q-adic ring. At order q^n, the new coefficient occurs only as $(\Omega + 1)V_n q^n$; coefficient extraction gives Equation (9.12). □

9.4 Universal first coefficients

Let

$$B = \Lambda - b = \frac{\Omega}{p}, \qquad D_2 = \frac{1}{2}\left[p(p-1)B + 2p - 1\right]. \tag{9.14}$$

Then

$$V_1 = -\frac{A_0(\Lambda)}{a(\Omega+1)}, \tag{9.15}$$

and

$$\begin{aligned} V_2 = -\frac{1}{\Omega+1}\Bigg[&D_2 V_1^2 + \frac{1}{a}\left(A_0'(\Lambda) + (p-\sigma)A_0(\Lambda)\right)V_1 \\ &+ \frac{1}{a}A_1(\Lambda)\Bigg]. \end{aligned} \tag{9.16}$$

These formulas show explicitly how three effects combine: curvature of the dominant phase, variation of the slow coefficient, and the next direct asymptotic block.

9.5 Analytic promotion by residual control

Formal algebra determines coefficients, but asymptotic validity is an analytic statement. The following elementary theorem is the bridge.

Theorem 9.3 (Residual-to-inverse error)**.** *Let F be continuously differentiable and strictly monotone on an interval containing the exact inverse point $x_* = F^{-1}(y)$ and an approximation x_N. Put*

$$R_N = F(x_N) - y.$$

Then, for some ξ between x_N and x_*,

$$x_N - x_* = \frac{R_N}{F'(\xi)}. \tag{9.17}$$

If F has the leading form in Equation (9.1), then

$$F'(\rho) \sim a\rho^{p-1}(\Omega + 1). \tag{9.18}$$

Hence a normalized residual of order q^{N+1} produces an inverse error of the same order as the first omitted transseries term.

Proof. The mean-value theorem gives

$$F(x_N) - F(x_*) = F'(\xi)(x_N - x_*),$$

which is Equation (9.17). Differentiating $ax^p(\log x - b)$ gives

$$ax^{p-1}\big(p(\log x - b) + 1\big),$$

and evaluation at $x = \rho$ gives Equation (9.18). □

This theorem supplies a useful certification workflow: derive the formal coefficients, substitute the truncation into the original asymptotic expansion, estimate the residual, and divide by the derivative scale. It also makes clear why a small residual in the function value can correspond to an even smaller error in the inverse when F' is large.

9.6 Newton acceleration in the transseries ring

Proposition 9.4 (Quadratic improvement)**.** *Suppose $x_0 = \rho(1+v_0)$ has normalized error $v_0 - v = O(q^m)$, with $m \geq 1$, and the linear factor $\Omega + 1$ is a unit. One formal Newton step*

$$x_1 = x_0 - \frac{F(x_0) - y}{F'(x_0)} \tag{9.19}$$

then has normalized error $v_1 - v = O(q^{2m})$.

Proof. Taylor expansion around the exact formal solution gives

$$F(x_0) - y = F'(x)(x_0 - x) + O((x_0 - x)^2).$$

Because the normalized derivative is a unit, division is permitted in the coefficient ring. The linear error cancels in Equation (9.19), leaving a quadratic remainder. □

Newton's method is therefore not merely a numerical afterthought. It is an algebraic accelerator that can double the number of correct q-adic blocks at each step.

9.7 Nonlattice supports and Hahn-field extension

The lattice $q^n = \rho^{-n\sigma}$ is convenient but not essential. If the direct expansion contains powers with exponents in a well-ordered additive semigroup $\mathcal{M} \subset \mathbb{R}_{\geq 0}$, one replaces $\mathcal{S}[[q]]$ by the corresponding Hahn series ring

$$\mathcal{S}((\mathcal{M})).$$

The same proof works coefficient by coefficient: at every monomial, the new unknown appears multiplied by $\Omega + 1$, while all remaining terms depend on earlier monomials. Nested logarithms or additional exponential scales are handled by enlarging the slow coefficient field and the monomial group, provided the support remains well ordered.

10 Coordinate normal forms beyond the K-function

10.1 Why normal forms matter

The canonical theorem applies directly to Equation (9.1), but it may not give the most efficient coordinates. A lower-order term such as $\beta \log x$ creates an inverse displacement much larger than subsequent algebraic corrections. Lemma 7.1 shows that this entire sector can be absorbed by an algebraic change of variable $T = x^p + \beta/a$. The resulting leading equation is again inverted by W, now in T rather than x.

Suppose

$$F(x) = ax^p(\log x - b) + \beta \log x + \gamma + O(x^{-p}). \tag{10.1}$$

Set

$$T = x^p + \frac{\beta}{a}, \qquad \gamma_* = \gamma - \beta\left(\frac{1}{p} - b\right). \tag{10.2}$$

Then

$$F(x) = \frac{a}{p} T(\log T - pb) + \gamma_* + O(T^{-1}). \tag{10.3}$$

The exact dominant inverse is

$$T_0(y) = \frac{p(y - \gamma_*)/a}{W_k\big((p/a)\mathrm{e}^{-pb}(y - \gamma_*)\big)}. \tag{10.4}$$

For the K-function, $p = 2$, $a = 1/2$, $b = 1/2$, and $\gamma_* = C$, reproducing Equation (7.13).

10.2 A normal-form algorithm

For more complicated direct expansions, one can seek

$$T = x^p + \alpha_0 + \alpha_1 x^{-\sigma} + \alpha_2 x^{-2\sigma} + \cdots \tag{10.5}$$

so that $F(x)$, expressed as a function of T, contains as few slow logarithmic corrections as possible. Coefficients are chosen successively by matching the unwanted terms. This is analogous to normal-form reduction in dynamical systems: a coordinate change removes nonresonant terms, while terms that cannot be removed become invariants of the asymptotic class.

For K, the single constant shift in r^2 already removes the only logarithmic correction, and the remaining expansion is a pure inverse-power series in T. This unusually clean outcome is the source of the rapid convergence observed in Table 2.

11 Complex inverse branches

11.1 Two independent branch indices

Let Z tend to infinity in a complex sector. Choose a branch of $\operatorname{Log} K(z)$ in a corresponding large-z sector and a branch of $\operatorname{Log} Z$. The equation $K(z) = Z$ lifts locally to

$$\kappa(z) = Y_m, \qquad Y_m = \operatorname{Log} Z + 2\pi \mathrm{i} m, \qquad m \in \mathbb{Z}. \tag{11.1}$$

The integer m records the logarithmic sheet of the outer exponential. The dominant phase introduces a second index, the Lambert branch W_k. Define

$$w_{m,k} = W_k\left(\frac{4Y_m}{\mathrm{e}}\right), \qquad \rho_{m,k} = \exp\left(\frac{w_{m,k}+1}{2}\right), \qquad L_{m,k} = \frac{w_{m,k}+1}{2}. \tag{11.2}$$

Whenever $\rho_{m,k}$ lies in the selected asymptotic sector and the logarithms are compatible, the same coefficient recurrence gives

$$z_{m,k}(Z) \sim \frac{1}{2} + \rho_{m,k} + \sum_{n\geq 1} V_n(L_{m,k})\rho_{m,k}^{1-2n}. \tag{11.3}$$

Remark 11.1. Formula Equation (11.3) is a sectorial branch construction, not a global enumeration theorem. Different pairs (m, k) may describe the same analytic branch after continuation, and some pairs may violate the chosen logarithmic sector. A complete quotient by these identifications requires a separate global study of the inverse Riemann surface.

11.2 The resonance or critical case

In the general theory, ordinary implicit inversion fails when

$$\Omega + 1 = 0. \tag{11.4}$$

This is exactly the point where the derivative of the dominant map vanishes. Since $\Omega = W_k(\cdot)$, it corresponds to the Lambert branch point $-\mathrm{e}^{-1}$. Near such a point, the correct local inverse scale is Puiseux-like, beginning with a square root, rather than an ordinary q-series. On the principal positive branch, $\Omega > 0$, so this resonance never occurs.

11.3 The accelerated complex form

The quadratic coordinate also extends sectorially. With

$$\eta_m = Y_m - C, \qquad \omega_{m,k} = W_k\left(\frac{4\eta_m}{\mathrm{e}}\right), \qquad \tau_{m,k} = \frac{4\eta_m}{\omega_{m,k}}, \tag{11.5}$$

the expansion is

$$z_{m,k}(Z) \sim \frac{1}{2} + \left[\frac{1}{12} + \tau_{m,k}\left(1 + \sum_{n\geq 2} U_n(S_{m,k})\tau_{m,k}^{-n}\right)\right]^{1/2}, \tag{11.6}$$

where $S_{m,k} = \operatorname{Log} \tau_{m,k}$, and the square-root branch must be chosen consistently with the target sector.

12 Exponentially small sectors and their inversion

12.1 What lies beyond the algebraic series

The coefficients in Equation (4.2) grow factorially. The optimally truncated remainder is exponentially small, and the Barnes G-function has a well-developed exponentially improved asymptotic theory with Stokes smoothing [7]. Through Equation (2.3), the same phenomenon enters the K-function. The explicit form depends on the complex sector: Stokes lines and terminant functions cannot be represented by a single globally valid power series.

For inversion, however, the leading transport mechanism is universal.

Theorem 12.1 (Transport of a first exponential sector)**.** *Suppose, in a fixed sector,*

$$F(x) = F_{\mathrm{alg}}(x) + S(x)\mathrm{e}^{-\lambda x} + \textit{smaller exponential sectors}, \qquad \Re(\lambda x) > 0, \tag{12.1}$$

and let $x_{\mathrm{alg}}(y)$ solve $F_{\mathrm{alg}}(x) = y$. If $F'_{\mathrm{alg}}(x_{\mathrm{alg}}) \neq 0$, then the inverse has the first exponential correction

$$F^{-1}(y) = x_{\mathrm{alg}}(y) - \frac{S(x_{\mathrm{alg}}(y))}{F'_{\mathrm{alg}}(x_{\mathrm{alg}}(y))}\mathrm{e}^{-\lambda x_{\mathrm{alg}}(y)} + \cdots . \tag{12.2}$$

Proof. Write $x = x_{\mathrm{alg}} + \delta$ and substitute into Equation (12.1). Taylor expansion gives

$$0 = F'_{\mathrm{alg}}(x_{\mathrm{alg}})\delta + S(x_{\mathrm{alg}})\mathrm{e}^{-\lambda x_{\mathrm{alg}}} + \text{quadratically or exponentially smaller terms.}$$

Solving the leading linear equation yields Equation (12.2). □

For the K-function, $F'_{\mathrm{alg}} = \kappa' \sim r\log r$. Thus a direct exponential contribution of scale $\mathrm{e}^{-2\pi r}$ is transported to an inverse contribution of scale

$$\frac{\mathrm{e}^{-2\pi\rho}}{\rho L} \tag{12.3}$$

up to its sector-dependent prefactor. Higher exponential sectors interact through the usual transseries convolution rules.

13 Applications to related functions

13.1 Generalized hyperfactorials

For $s > -1$, consider the discrete generalized hyperfactorial

$$H_s(n) = \prod_{j=1}^{n} j^{j^s}. \tag{13.1}$$

Euler–Maclaurin summation applied to $\sum_{j\le n} j^s \log j$ begins with

$$\log H_s(n) \sim \frac{n^{s+1}}{s+1}\log n - \frac{n^{s+1}}{(s+1)^2} + \frac{1}{2}n^s\log n + \cdots . \tag{13.2}$$

Ignoring for the moment the boundary correction and choosing a smooth interpolation, the dominant parameters are

$$p = s+1, \qquad a = \frac{1}{s+1}, \qquad b = \frac{1}{s+1}. \tag{13.3}$$

Therefore the first Lambert-normalized scale is

$$\rho_s(Y) = \left[\frac{(s+1)^2 Y}{W_0((s+1)^2 Y/\mathrm{e})}\right]^{1/(s+1)}. \tag{13.4}$$

The lower-order Euler–Maclaurin terms determine the centering and the coefficient functions A_m. Formula Equation (9.12) then constructs the inverse to all orders. The ordinary hyperfactorial is $s = 1$, for which Equation (13.4) reduces to $2\sqrt{Y/W_0(4Y/\mathrm{e})}$.

13.2 The inverse gamma scale

Stirling's formula for $\log \Gamma(x)$ has dominant part

$$x(\log x - 1),$$

corresponding to $a = 1$, $p = 1$, $b = 1$. The leading inverse scale is therefore

$$\rho(Y) = \frac{Y}{W_0(Y/\mathrm{e})}. \tag{13.5}$$

The familiar inverse-factorial and inverse-gamma expansions are thus the $p = 1$ member of the same theory, while the K-function is the $p = 2$ member.

13.3 Barnes-type and multiple-gamma functions

Logarithms of Barnes multiple gamma functions have leading terms of the form $x^p \log x$ with integer p. After removal of polynomial terms and appropriate centering, their inverse functions fit Equation (9.1). The dominant W formula Equation (9.6) is therefore universal across the hierarchy; only the slow coefficient field and the exponent lattice change. In higher order, several polynomial-logarithmic terms may need to be absorbed by a more elaborate coordinate normal form before the residual recurrence becomes sparse.

14 Algorithms and reproducibility

14.1 Coefficient-generation algorithm

The canonical K coefficients can be generated without expanding the entire expression at once.

Algorithm 14.1 (Triangular coefficient generation)**.** *Given a target order N:*

1. *Compute $c_1, \ldots, c_{N-1}$ from* Equation (4.3).
2. *Initialize $v = 0$.*
3. *For $n = 1, \ldots, N$, truncate every power series in* Equation (6.5) *modulo q^{n+1}, extract the coefficient of q^n, divide its known part by $-L$, and append $V_n q^n$ to v.*
4. *Optionally simplify V_n as a Laurent polynomial in L and C.*
5. *Verify by substituting the resulting v back into* Equation (6.5) *and checking that the residual is $O(q^{N+1})$.*

The accelerated coefficients are generated identically from Equation (7.17), except that the unknown series starts at v^2 and the divisor is S.

14.2 Stable numerical evaluation

For very large Z, one should never form Z explicitly. The natural input is $Y = \log Z$. Compute $W_0(4Y/\mathrm{e})$ or $W_0(4(Y - C)/\mathrm{e})$, evaluate the selected transseries truncation, and only then use the exact special functions if a Newton correction is desired. This avoids overflow and respects the asymptotic geometry of the problem.

A residual provides an a posteriori error estimate:

$$\widehat{\epsilon} = \frac{\kappa(x_{\mathrm{app}}) - Y}{\kappa'(x_{\mathrm{app}})}. \tag{14.1}$$

When the approximation is already in the asymptotic basin, $x_{\mathrm{app}} - x_* \approx \widehat{\epsilon}$, and subtracting $\widehat{\epsilon}$ is one Newton step.

14.3 Reference Python code

The following script reproduces the numerical tables. It uses `mpmath` for high-precision gamma, Barnes G, and Lambert W evaluations.

Listing 1: Numerical verification of the inverse K-function transseries.

```
import mpmath as mp

mp.mp.dps = 80

# log(A) = 1/12 - zeta'(-1), and C = log(A) - 1/8.
logA = mp.mpf(1) / 12 - mp.diff(lambda s: mp.zeta(s), -1)
C = logA - mp.mpf(1) / 8


def logK(x):
    """Real log K(x), using K(x) G(x) = Gamma(x)^(x-1)."""
    x = mp.mpf(x)
    return (x - 1) * mp.loggamma(x) - mp.log(mp.barnesg(x))


def dlogK(x):
    """Derivative of log K(x)."""
    x = mp.mpf(x)
    return (mp.loggamma(x) + x - mp.mpf("0.5")
            - mp.log(2 * mp.pi) / 2)


def inverse_logK(y):
    """Solve log K(x) = y on x >= 2 by Newton iteration."""
    y = mp.mpf(y)
    w = mp.lambertw(4 * y / mp.e)
    x = mp.mpf("0.5") + 2 * mp.sqrt(y / w)
    for _ in range(30):
        step = (logK(x) - y) / dlogK(x)
        x -= step
        if abs(step) < mp.mpf("1e-70"):
            break
    return x


def canonical(y):
    """Return X_0, X_1, X_2, X_3."""
    y = mp.mpf(y)
    w = mp.lambertw(4 * y / mp.e)
    rho = 2 * mp.sqrt(y / w)
    L = mp.log(rho)
    E = C - L / 24
    c1 = -mp.mpf(7) / 5760
    c2 = mp.mpf(31) / 161280

    V1 = -E / L
    V2 = -((L + 1) * V1**2 / 2 - V1 / 24 + c1) / L
    V3 = -((L + 1) * V1 * V2 + V1**3 / 6
           - (V2 - V1**2 / 2) / 24
           - 2 * c1 * V1 + c2) / L

```

```
    X0 = mp.mpf("0.5") + rho
    X1 = X0 + V1 / rho
    X2 = X1 + V2 / rho**3
    X3 = X2 + V3 / rho**5
    return X0, X1, X2, X3


def accelerated(y):
    """Return A_0, A_2, A_3, A_4, A_5."""
    y = mp.mpf(y)
    eta = y - C
    omega = mp.lambertw(4 * eta / mp.e)
    tau = 4 * eta / omega
    S = mp.log(tau)

    U2 = 1 / (120 * S)
    U3 = -mp.mpf(31) / (22680 * S)
    U4 = ((1030 * S**2 - 126 * S - 63)
          / (mp.mpf(1814400) * S**3))
    U5 = -((16110 * S**2 - 1023 * S - 341)
           / (mp.mpf(29937600) * S**3))

    series = [
        mp.mpf(1),
        1 + U2 / tau**2,
        1 + U2 / tau**2 + U3 / tau**3,
        1 + U2 / tau**2 + U3 / tau**3 + U4 / tau**4,
        1 + U2 / tau**2 + U3 / tau**3
          + U4 / tau**4 + U5 / tau**5,
    ]
    return tuple(mp.mpf("0.5")
                 + mp.sqrt(mp.mpf(1) / 12 + tau * s)
                 for s in series)


for y in [10, 100, 1000, 10000, mp.mpf("1e6")]:
    exact = inverse_logK(y)
    print("Y =", y)
    print("x =", mp.nstr(exact, 30))
    print("canonical errors:",
          [mp.nstr(x - exact, 12) for x in canonical(y)])
    print("accelerated errors:",
          [mp.nstr(x - exact, 12) for x in accelerated(y)])
```

15 Further deductions and research directions

15.1 Late-term growth of the inverse coefficients

The direct coefficients c_n inherit factorial growth from Bernoulli numbers. The triangular recurrence then forces factorial growth of the inverse coefficient sequence as well. A detailed late-term analysis should identify the Borel singularities and Stokes constants of the inverse directly from those of $\log K$.

Conjecture 15.1 (Inherited resurgent singularities)**.** *After the natural normalization by the derivative of the dominant phase, the Borel singularities controlling the late terms of the inverse K transseries occur at the same action values as those of the centered direct K expansion. In particular, the leading action* *magnitude on the positive ray is $2\pi\rho$, and the inverse Stokes constants are obtained from the direct constants by the transport law in Theorem 12.1, followed by nonlinear convolution for higher sectors.*

This is strongly suggested by general resurgent inversion principles, but a complete proof would require a sectorially normalized transseries for $\log K$ and careful control of composition in the relevant resurgent algebra.

15.2 Optimality of the quadratic coordinate

Among transformations of the form $T = r^2 + \alpha$, the choice $\alpha = -1/12$ is uniquely forced by cancellation of the logarithmic correction. A broader optimization problem asks whether a full formal coordinate

$$T = r^2 - \frac{1}{12} + \alpha_1 r^{-2} + \alpha_2 r^{-4} + \cdots$$

can remove arbitrarily many of the $d_n T^{-n}$ terms. Formally, many terms can be shifted between the coordinate and the residual series, but useful normal forms should also preserve simplicity, invertibility, and sectorial analyticity. The best compromise may be characterized by a canonical resurgence or Borel-normalization condition.

15.3 Uniform asymptotics near the Lambert branch point

The ordinary recurrence divides by $\Omega + 1$, so it is not uniform near $\Omega = -1$. A separate double-scaling theory should combine the square-root expansion of W near $-\mathrm{e}^{-1}$ with the lower-order transseries of F. Such a theory would describe the merger of inverse branches and is relevant to global complex continuation.

15.4 Formalization prospects

The algebraic core is well suited to proof-assistant formalization:

- define the q-adic coefficient ring with a distinguished slow parameter;
- formalize the implicit recurrence and uniqueness proof;
- verify coefficient identities such as Equations (6.7) to (6.9);
- separate these formal results from analytic theorems about Poincare remainders and monotonicity.

The analytic promotion theorem can then be connected to existing formalized real analysis through explicit residual bounds.

15.5 Global inverse geometry

A full account of the inverse K-function should determine:

1. the critical points and critical values of K;
2. the monodromy relating the (m, k) asymptotic labels;
3. the sectors in which each asymptotic branch is dominant;
4. the Stokes graph for exponentially improved inverse expansions.

The formulas in Section 11 provide asymptotic local charts for this future Riemann-surface description.

16 Conclusion

The large inverse of the K-function becomes systematic once its dominant power–logarithmic phase is inverted exactly. The unique centering $x = r + \frac{1}{2}$ reveals an even asymptotic expansion, and the Lambert normalizer

$$\rho = 2\sqrt{\frac{\log Z}{W_0(4\log Z/\mathrm{e})}}$$

turns inversion into a triangular series in ρ^{-2}. This yields an all-orders formula with explicitly recursive coefficients.

The further coordinate

$$T = x^2 - x + \frac{1}{6}$$

is not an ad hoc numerical trick. It is the normal-form transformation that absorbs the logarithmic correction dictated by the centered expansion. In that coordinate, the leading map is $\frac{1}{4}T(\log T - 1) + C$, its inverse is again exact in terms of W, and the remaining correction begins two powers later in the relative scale. The result is both conceptually cleaner and numerically superior.

More generally, every transseries with dominant part $ax^p(\log x - b)$ has an exact W-adapted leading inverse. Once this scale is chosen, formal inversion is an implicit-function problem over a slow coefficient field; residual estimates promote formal identities to analytic asymptotics; Newton iteration accelerates the expansion; and exponentially small direct sectors are transported by division by the dominant derivative. The K-function therefore serves as a particularly transparent model for a large family of asymptotic inverse problems.

A Coefficient derivations

A.1 Centered coefficients

Starting from Equation (4.4), the coefficient of r^{-2n-1} in $\kappa'(r + \frac{1}{2})$ is

$$\frac{B_{2n+2}(1/2)}{(2n+1)(2n+2)}.$$

Integration gives

$$-\frac{B_{2n+2}(1/2)}{(2n)(2n+1)(2n+2)}r^{-2n},$$

which is Equation (4.3). The identity

$$B_{2n+2}(1/2) = \left(2^{-2n-1} - 1\right) B_{2n+2}$$

may be used to express all coefficients in terms of ordinary Bernoulli numbers.

A.2 Quadratic-normal-form coefficients

From the proof of Theorem 7.2,

$$\kappa(x) - \Psi(T) - C = \frac{T}{4}\log\left(1 + \frac{a_0}{T}\right) - \frac{a_0}{4} + \sum_{m\geq 1} c_m (T + a_0)^{-m},$$

where $a_0 = 1/12$. The logarithmic part contributes to T^{-n}

$$\frac{(-1)^n a_0^{n+1}}{4(n+1)}.$$

Furthermore,

$$(T + a_0)^{-m} = T^{-m} \sum_{j\geq 0} (-1)^j \binom{m+j-1}{j} a_0^j T^{-j}.$$

Taking $j = n - m$ gives

$$(-1)^{n-m}\binom{n-1}{m-1}a_0^{n-m},$$

and summing over $m \le n$ proves Equation (7.9).

B A compact symbolic pseudocode

The following language-independent pseudocode emphasizes that only truncated series arithmetic is required.

Listing 2: Formal recurrence for the canonical coefficients.

```
INPUT: target order N, symbols L and C
q := formal small parameter
E := C - L/24
v := 0

for n from 1 to N:
    D := (1/2)*(1+v)^2*(L + log(1+v))
         - (1/4)*(1+v)^2 - L/2 + 1/4

    B := E - log(1+v)/24
    for m from 1 to n-1:
        c[m] := -BernoulliPolynomial(2*m+2, 1/2)
                / ((2*m)*(2*m+1)*(2*m+2))
        B := B + c[m]*q^m*(1+v)^(-2*m)

    known := coefficient(truncate(D + q*B, q^(n+1)), q^n)
    V[n] := simplify(-known/L)
    v := v + V[n]*q^n

OUTPUT: V[1], ..., V[N]
CHECK: truncate(D + q*B, q^(N+1)) == 0
```

C Bibliographic notes

The K-function definitions and functional identities may be found in standard special-function references and in the concise summary [6]. Barnes G asymptotics and the Glaisher constant are recorded in [4]; rigorous error bounds and exponential improvement are developed in [7]. Lambert W and its large argument expansion are treated in [3, 5]. General transseries background may be found in [10], while classical asymptotic error methods are covered by [8]. Recent hyperfactorial refinement literature includes [11].

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